

The Cartesian Plane

Name: _____

Date: _____

Year 6 Maths — Space

ACARA v9.0: AC9M6SP02

Learning Intention

I am learning to locate points in the 4 quadrants of a Cartesian plane and describe changes to coordinates when a point moves.

Success Criteria

- I can plot a point using an (x, y) coordinate in any quadrant.
- I can identify which quadrant a coordinate is in.
- I can describe how coordinates change when a point moves.

Along the Hall, Up the Stairs

A coordinate (x, y) tells you: x = how far left/right from zero, y = how far up/down from zero. Positive and negative values place the point in one of 4 quadrants.

Worked Example

Point (3, -2): move 3 right, 2 down.
This point is in Quadrant 4 (positive x, negative y).

Part A — Warm Up

In the coordinate (5, 2), which number tells you how far up to go? _____

True or False: The point (-3, 4) is in Quadrant 2. _____

Name the point where the x-axis and y-axis cross. _____

Part B — Core Skills

Which quadrant is (-4, -1) in? _____

A point starts at (2, 5) and moves 3 units left. What is its new coordinate? _____

A point starts at (-1, -6) and moves 4 units up. What is its new coordinate? _____

Explain how you know a point is in Quadrant 3 without plotting it.

Word Bank

Cartesian plane • x-axis • y-axis • coordinate • quadrant • origin • ordered pair

Working Space — Plot and Label 4 Points, One in Each Quadrant

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Part C — Challenge: The Treasure Map

A Year 6 class is creating a treasure map on a Cartesian plane. The treasure starts buried at $(-2, 3)$. The map's key says: "Move the treasure 5 units right, then 6 units down to reveal its new hiding spot."

Q8. Calculate the treasure's new coordinate after following the map's instructions.

Q9. Identify which quadrant the treasure's new coordinate is in.

Q10. A second team says the treasure ends up in Quadrant 1 instead. Explain the mistake they most likely made in reading the map's instructions.

Answer Key — Part A

A1: 2 (the y-value).

A2: True — negative x, positive y is Quadrant 2.

A3: The origin $(0, 0)$.

Answer Key — Part B

B1: Quadrant 3 (negative x, negative y).

B2: $(-1, 5)$.

B3: $(-1, -2)$.

B4: In Quadrant 3, both the x-coordinate and y-coordinate are negative, so if both numbers in the ordered pair are negative, the point must be in Quadrant 3.

Answer Key — Part C

C8: $(3, -3)$ — moving 5 right from -2 gives 3 (x), and 6 down from 3 gives -3 (y).

C9: Quadrant 4 (positive x, negative y).

C10: They likely moved right and up instead of right and down, or misread the direction of the 'down' instruction, landing in Quadrant 1 (positive x, positive y) instead of correctly ending with a negative y-value in Quadrant 4.

I Can Checklist

☐ I can plot points in any quadrant.

☐ I can identify which quadrant a point is in.

☐ I can describe coordinate changes when a point moves.